Part D: Team Contract — DebateSpeak AI

Field	Description
Course	CS300 / CSC13002 — Introduction to Software Engineering
Document	**Part D: Team Contract & Collaboration Framework**
Score Allocation	Part D (5 Points)

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1. Team Roles and Responsibilities

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In accordance with course guidelines, **all 5 team members act as full-stack engineers** contributing to design, coding, testing, and documentation. To maintain accountability and prevent gaps, each member leads a dedicated functional subsystem:

Member Name & ID	Primary Lead Role	Full-Stack Responsibilities & Lead Deliverables
Nguyễn Bảo Minh Triết (ID: `24125047`)	**Frontend & WebRTC/Audio Lead (Triết)**	• Leads the Vite + React SPA architecture, Tailwind design system, and UI accessibility. • Implements the LiveKit Client SDK integration, audio controls, room lobby, and interactive report viewer. • Coordinates user flows with backend WebSocket endpoints.
Nguyễn Hồng Tấn Tài (ID: `24125078`)	**Backend & Real-Time Media Lead (Tài)**	• Leads the FastAPI core architecture and PostgreSQL database schema design (SQLAlchemy/Alembic). • Implements user authentication (Argon2id, rotating JWTs) and room lifecycle state machine. • Develops the LiveKit server token minting service and Redis pub/sub WebSocket hub.
Nguyễn Anh Khoa (ID: `24125058`)	**AI Audio & Speech Pipeline Lead (Khoa)**	• Leads the LiveKit audio track subscription worker (bot subscriber). • Implements Silero VAD for vocal activity segmentation and pause measurement. • Develops the streaming STT adapter pipeline (cloud streaming STT + local fallback engine), word timestamps, and filler-word tracking.
Trần Lê Anh Tuấn (ID: `24125107`)	**AI Orchestrator & Analysis Lead (Tuấn)**	• Leads the prompt engineering, LLM adapter interfaces, and Pydantic schemas for all AI agents. • Implements the Language Coach (grammar/vocab), Argument Coach (rebuttal/fallacies), and Debate Judge. • Builds the deterministic Consensus Engine and the regex/substring quote-grounding guard.
Nguyễn Đình Thiên Lộc (ID: `24125093`)	**DevOps & QA / Fact-Checking Lead & Release Custodian (Lộc)**	• Leads the Fact Investigator search retrieval pipeline (web search adapter) and Evidence Evaluator agent. • Serves as **Documentation Custodian & Git Release Manager** for Sprint 1 (PA1): bootstraps repository, manages branch merges, compiles LaTeX reports via `latexmk`, and packages release archives on behalf of the collaborative authoring team. • Manages the Docker Compose multi-container staging environment and Caddy automated TLS reverse proxy. • Establishes CI/CD pipelines (GitHub Actions), Playwright E2E automation, and golden-set evaluation scripts.

> **Note on Sprint 1 (PA1) Git Commit Log Attribution:**
> During Sprint 1 (project conception and documentation phase), all five team members collaborated extensively on drafting, cross-reviewing, and editing the required deliverables (as certified by the per-section attribution headers across all project documents). To prevent merge conflicts during rapid document iteration and ensure uniform Markdown/LaTeX formatting, **Nguyễn Đình Thiên Lộc acted as the central Documentation Custodian & Git Release Manager**, committing the consolidated and verified submissions to the remote repository. Individual member Git commit activity will take full effect starting in Sprint 2 (PA2) as code implementation begins on dedicated feature branches.

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2. Communication Plan

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- **Primary Communication Channels:**
  - **Facebook Messenger:** Primary daily communication channel (team group chat for daily coordination, instant messaging, announcements, urgent blocker alerts, and poll-based meeting scheduling).
  - **Google Meet:** Synchronous online team meetings, sprint ceremonies, and live pair-programming sessions (Meeting Portal Evidence).
  - **GitHub Discussions / PR Comments:** Formal code reviews, architectural debate, and technical RFCs.

- **Meeting Cadence & Scheduling:**
  - **Meeting Times:** Specific meeting times and dates are **flexible and to be decided** collaboratively prior to each sprint or milestone based on all 5 members' course schedules and academic availability (polled and confirmed via Messenger).
  - **Sprint Ceremonies & Syncs:** Arranged dynamically at key milestones (Sprint Planning at sprint launch, mid-sprint progress check-ins, and sprint review/retro demonstrations) to maintain momentum without rigid scheduling constraints.

- **Response Time Expectations:**
  - Standard workdays (Monday – Friday, 08:00 – 21:00): Acknowledge messages within **4 hours**.
  - Urgent blockers (`[BLOCKER]` prefix in Messenger): Acknowledge within **2 hours**.
  - Weekends: Acknowledge within **12 hours** unless notified of planned absence in advance.

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3. Work Schedule and Deadlines

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- **Incremental Release Schedule (5 Sprints mapped to PA1–PA5):**
  - **Sprint 1 (PA1 - Weeks 1–2):** *Conception & Setup* — Proposal (10 FGs), app survey, team contract, dev tooling, baseline monorepo, Docker Compose, and academic report.
  - **Sprint 2 (PA2 - Weeks 3–4):** *Planning & Foundation (Build 1)* — Project Plan, Vision Document, Spec Kit initialization in `/src`, and Build 1 (Auth + Room Lobby state machine skeleton).
  - **Sprint 3 (PA3 - Weeks 5–6):** *Use-Case Modeling & First Functional Slice (Build 2)* — Use-case specs & diagrams, Changes.md, Build 2 (End-to-End implementation of 1 FG: Room & Session Management), narrated video demo.
  - **Sprint 4 (PA4 - Weeks 7–8):** *Architecture & Second Implementation Slice (Build 3)* — C4 model (Context, Container, Component, Deployment), Build 3 (Implementation of 2 FGs: Live Audio + Streaming STT), narrated video demo.
  - **Sprint 5 (PA5 - Weeks 9–10):** *Comprehensive Testing, Demo & Hardening (Build 4 - Final Release)* — Test plan, refined test cases, golden-set evaluation, 15-minute live product demo of all 10 FGs, reflective report.

- **Contingency Protocol for Missed Deadlines:**
    - If a team member realizes a task will be delayed, they must notify the Team Leader on Messenger at least **48 hours before the sprint deadline**.
    - The team will hold an emergency triage meeting to redistribute subtasks or scope down non-critical features to preserve milestone delivery.

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4. Code and Documentation Standards

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- **Coding Conventions & Linters:**
    - **Python (Backend & Workers):** Strict PEP 8 standards enforced via **Ruff** (line length 100) and static type checking via **MyPy**.
    - **TypeScript (Frontend):** Enforced via **ESLint** and **Prettier** with strict TypeScript compiler checks (`noImplicitAny: true`).

- **Git Branching Strategy:**
    - Git Flow model: `main` (production-ready, tagged releases), `develop` (staging integration), and short-lived feature branches (`feat/room-timer` , `fix/vad-segmentation`).
    - **Never commit directly to `main` or `develop` .** All changes require a Pull Request.

- **Code Review & Testing Guidelines:**
    - Every PR must have at least **one approving review** from a designated peer reviewer before merging.
    - PRs must pass automated CI checks (linters, unit tests) and maintain ≥70% test coverage on core business logic.

- **Documentation Standards:**
    - All documentation is written in clear, concise English using GitHub-flavored Markdown.
    - Architectural flows must use Mermaid diagrams.

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5. Accountability, Performance, and Consequences

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- **Contribution Measurement Criteria:**
    - Completion of assigned Jira tasks within the committed sprint timeline.
    - Meaningful, consistent Git commit activity (measured via commit frequency and pull request contributions; no last-minute code dumps).
    - Punctual attendance and active participation in scheduled Scrum ceremonies.

- **Handling Underperformance:**

    - **Step 1 (Informal 1-on-1):** The Project Manager speaks privately with the struggling member to understand challenges, offer technical pair-programming support, or rebalance workload.
    - **Step 2 (Formal Team Notice):** If inactivity or unresponsiveness continues for more than 48 hours without justification, a written notice is posted in the private team log outlining required remediation actions within 3 business days.
    - **Step 3 (Consequence & Escalation):** If underperformance persists, the team will formally reflect the individual's contribution accurately in the peer evaluation form and escalate the issue to the course Teaching Assistant (TA) or Instructor.

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6. Decision-Making & Conflict Resolution

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- **Decision-Making Protocol:**
    - Architectural and technical decisions are made through open discussion and **consensus whenever possible**.
    - If consensus cannot be reached, the team takes a **majority vote (3 out of 5 votes)**.
    - In the event of a tie or time-critical impasse on schedule trade-offs, the **Project Manager holds the casting vote**.

- **Conflict Resolution Framework:**
    - **Level 1 (Internal Discussion):** Disputing parties discuss the issue professionally in a dedicated 1-on-1 meeting focusing strictly on objective code quality, project requirements, and data.
    - **Level 2 (Team Mediation):** If unresolved, the full team mediates the discussion during the next Scrum meeting.
    - **Level 3 (Instructor Escalation):** If a personal or structural dispute threatens project progress, the Project Manager contacts the TA/Instructor for academic mediation.

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7. Contract Review and Update Process

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- This team contract will be reviewed at the conclusion of each sprint during the **Sprint Retrospective**.
- Any team member may propose amendments to communication protocols, workload distribution, or tooling.
- Amendments take effect upon unanimous agreement of all 5 team members and will be committed to `docs/management/TEAM_CONTRACT.md` with an updated revision date.